

MATH3026/4022: TIME SERIES ANALYSIS/TIME SERIES AND FORECASTING

DETAILS REGARDING NON-EXAMINABLE MATERIAL AND SUPPLIED FORMULAE SPRING SEMESTER 2025-2026

This document contains information on non-examinable parts of MATH3026/4022 Lecture Slides and information on formulae that would be supplied if necessary.

A separate formula sheet is not included with the MATH3026/4022 examination, however, there are some results/formulae that students do not need to learn because these would be given, within a question, if needed.

Information on non-examinable material (or formulae that would be given), grouped by the lecture in which this occurs, is outlined in the table below. It should be assumed that anything not mentioned below could potentially appear in the MATH3026/4022 examination paper.

Note: This document covers the 2025-2026 Spring Semester examination and August 2026 resit only. Any information below may be revised in subsequent years.

Generally, students do not need to learn specific probability density/mass functions (e.g. multivariate normal). If required, these would be supplied within a question.

Students will not be asked to write down any R code. However, you may be asked to interpret R output (e.g. relevant plots, parameter estimates, P-values etc.).

Lecture	Information
Lecture 1	Non-examinable: Slide 29 – definition of a probability space.
Lecture 2	All material is examinable.
Lecture 3	All material is examinable.
Lecture 4	You do not need to learn by heart the stationarity conditions on Slide 38. However, you might be asked to derive some/all of these or work with them if stated (as in Exercises Sheet 2, Question 5).
Lecture 5	All material is examinable.
Lecture 6	Non-examinable: Recursive calculation of the PACF (slides 18-22). Formulae for c_0 and/or c_h (Slide 33) would be supplied (within a question) if needed. Non-examinable: Recursive calculation (Slide 37). Bartlett's formulae for the variance and correlation (Slides 38 and 39) would be supplied if needed.

Lecture 7	Students may be required to write likelihood functions and determine score equations. Probability density functions would be supplied in this case.
Lecture 8	Students may be required to write likelihood functions and determine score equations. Probability density functions would be supplied in this case. Formulae for c_0 and/or c_h would be supplied (within a question) if needed.
Lecture 9	You would not be required to evaluate information matrices (as in Slides 13-14) but you should know the definitions. Formulae for the test statistics on Slides 22 and 24 need not be known. However, you should know how to carry out tests and know the distribution of test statistics. You do not need to know how to write R commands (as in Slides 26, 28, 29). However, you may be asked to interpret output or plots from R. You don't need to learn the formula for AIC_c but you should know formulae for AIC and BIC.
Lecture 10	You do not need to know how to write R commands (as in Slide 41). However, you may be asked to interpret output or plots from R.
Lecture 11	All material is examinable.
Lecture 12	For "Forecasting in R" you may be asked to interpret R output (e.g. fitted model summaries or plots) but you would not be asked to write any R code in an answer.
Lecture 13	The derivation on Slides 32 and 33 is not examinable.
Lecture 14	You do not need to learn the formulae for the periodogram on Slides 15-16, but you may be asked to interpret/comment on a periodogram.
Lecture 15	All material is examinable.